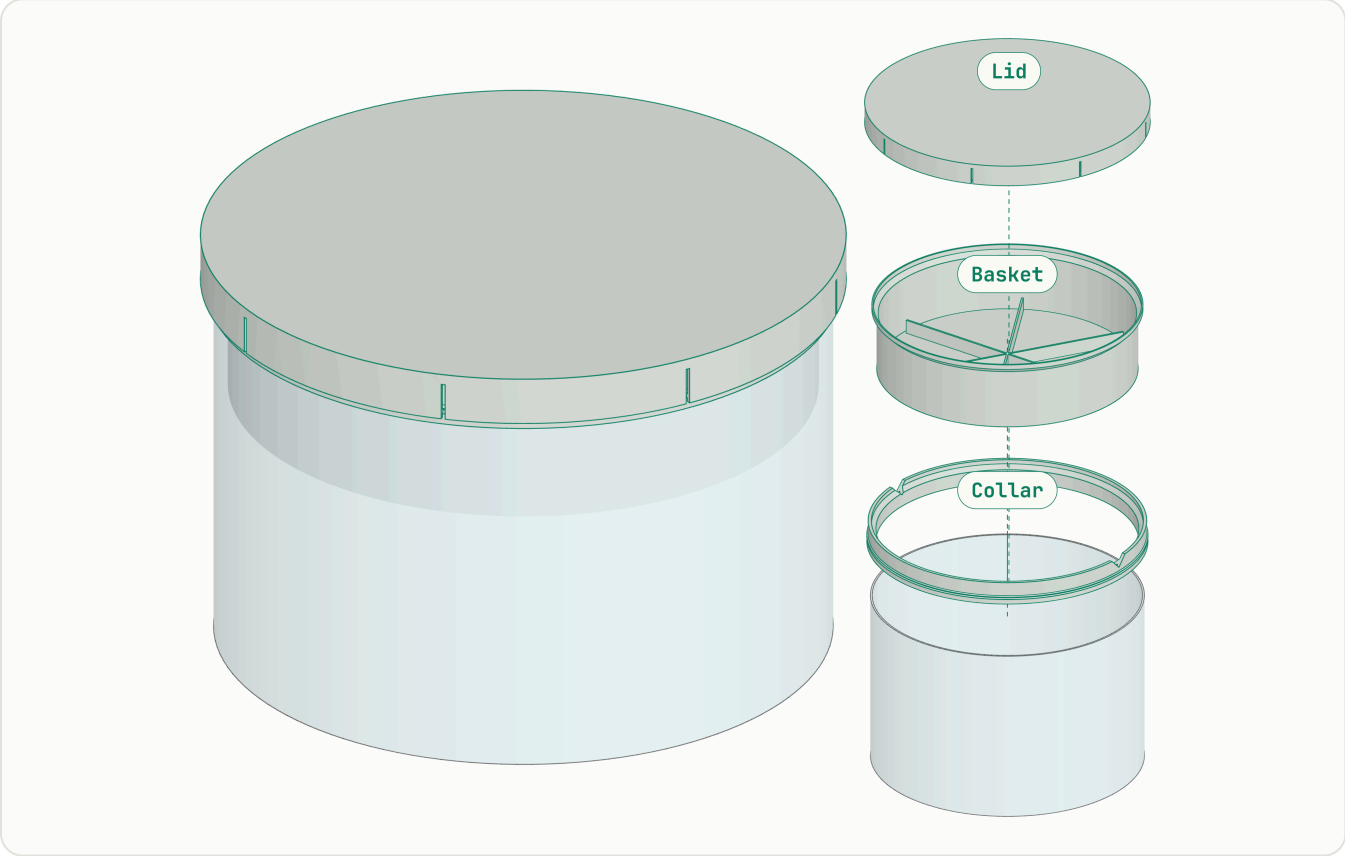
PINK STUFF

A desiccant basket for the bottom half of a pink antifreeze jug

Cut a West Marine -50°F antifreeze gallon jug in half with the cutting ring. A collar goes over the cut edge. A basket of desiccant pellets hangs in the collar. A lid snaps onto the collar and keeps the pellets in. The pellets take water from the air. The brine drips through the floor of the basket and collects in the jug, below the basket.



Left: the assembly on the jug. Right: the parts. From the bottom: the jug, the collar, the basket, the lid.

Measured and estimated. The model uses a 152.8 mm diameter, from a 480 mm circumference on a sample jug. The 1.0 mm wall is an estimate. Print the fit test arc first. `MEASURE.md` lists the measurements to take.

- ☐ White PLA
- 62 g and 1 h 40 min for each jug
- cutting ring 55 g, once
- No supports
- One M2.5 x 12 screw

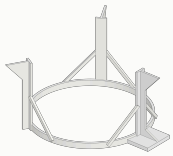
Printed parts

	File	Size (mm)	Each	Time each	Count	Filament	Time	Cost
	stl/jig.stl	191 × 194 × 96	44 g	1 h 56 min	1	44 g	1 h 56 min	$0.89
	stl/jig_cart.stl	13 × 48 × 29	11 g	23 min	1	11 g	23 min	$0.21
	stl/fit_arc.stl	9 × 50 × 13	2 g	6 min	1	2 g	6 min	$0.05
	stl/collar.stl	157 × 157 × 13	22 g	35 min	1	22 g	35 min	$0.44
	stl/basket.stl	151 × 151 × 40	26 g	38 min	1	26 g	38 min	$0.52
	stl/lid.stl	159 × 162 × 12	14 g	28 min	1	14 g	28 min	$0.28
	Total					**120 g**	**4 h 05 min**	**$2.39**

Filament and print time from the Printdeck slice: the Printdeck slicer with the TAZ Workhorse SE profile (pla-0.30-fast, faster speeds, white PLA), the same slice as printdeck.beetlebug.org. Cost at $19.99/kg. The counts are for 1 jugs.

Print

The STL files are in their print orientation. Load them, do not turn them.



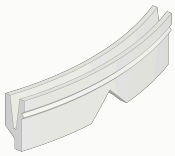
jig

support check 0 g
Upside down, the ring top on the bed. Print it once for all the jugs.



jig cart

support check 0 g
On its back end, the open end of the blade slot on the bed.



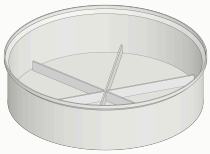
fit arc

support check 0 g
Top face on the bed. Print this first. It tests the groove and the diameter on the jug.



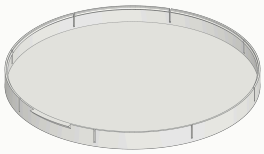
collar

support check 0 g
Top face on the bed (upside down). The groove opens to the top.



basket

support check 0 g
Floor on the bed. The rim flares out at 45°.



lid

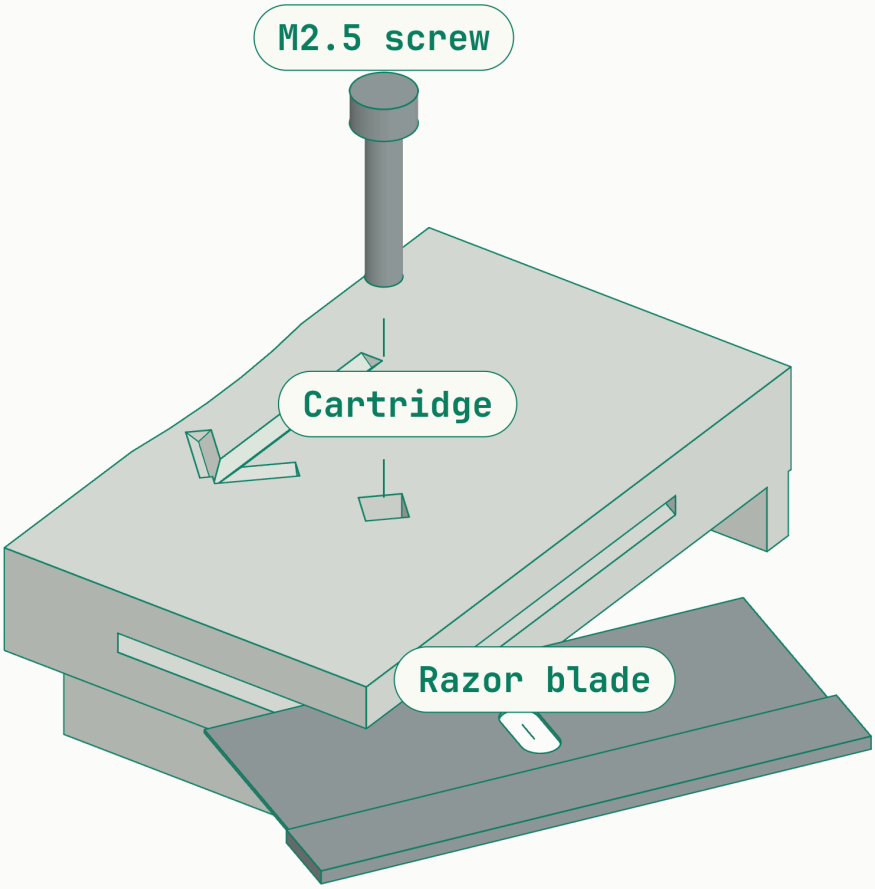
support check 0 g
Top face on the bed (upside down). The skirt goes up.

Settings (LulzBot TAZ Workhorse)

Printdeck: `taz-workhorse-se/pla-0.30-fast`, roll 5, with the part settings in printdeck.yml. Or in Cura: run

1 Load the blade

cartridge | razor blade | 2 mm hex key

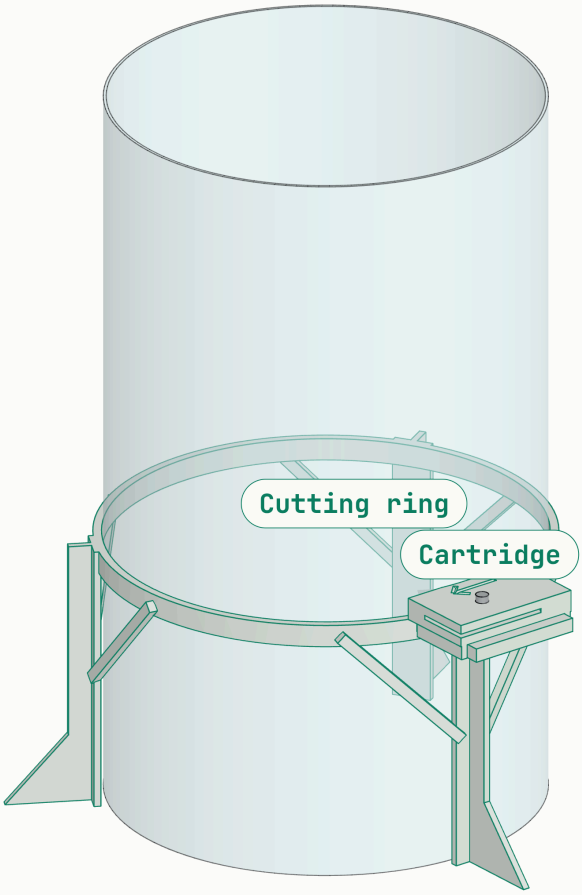


1. Use a single-edge razor blade with a folded steel spine: 1.5 in long, 11/16 to 3/4 in wide, 0.009 in thick.
2. Hold the blade by the spine. Point the edge away from you.
3. Find the back end of the cartridge: the slot is open at the back and at one side. The thin slot is for the blade, the thick part of the slot is for the spine.
4. Slide the blade into the open slot, edge first, flat, with the spine at the back. The slot is at an angle: follow it.
5. Push the spine in with the short end of the hex key until it stops. The spine stops at the step in the slot.
6. Look at the front of the cartridge (the curved end). The blade is at 30°. Its tip comes out about 10 mm.
7. Look down the screw hole. The hole must be clear behind the spine. If the spine covers it, push the blade in more.

Note. The spine is thicker than the blade slot, so the blade goes in only one way: edge first, toward the jug. To take the blade out, remove the screw. Hold the cartridge over a table, back end down. The blade slides out. Tap the cartridge if the blade does not move.

2 Cut the jug

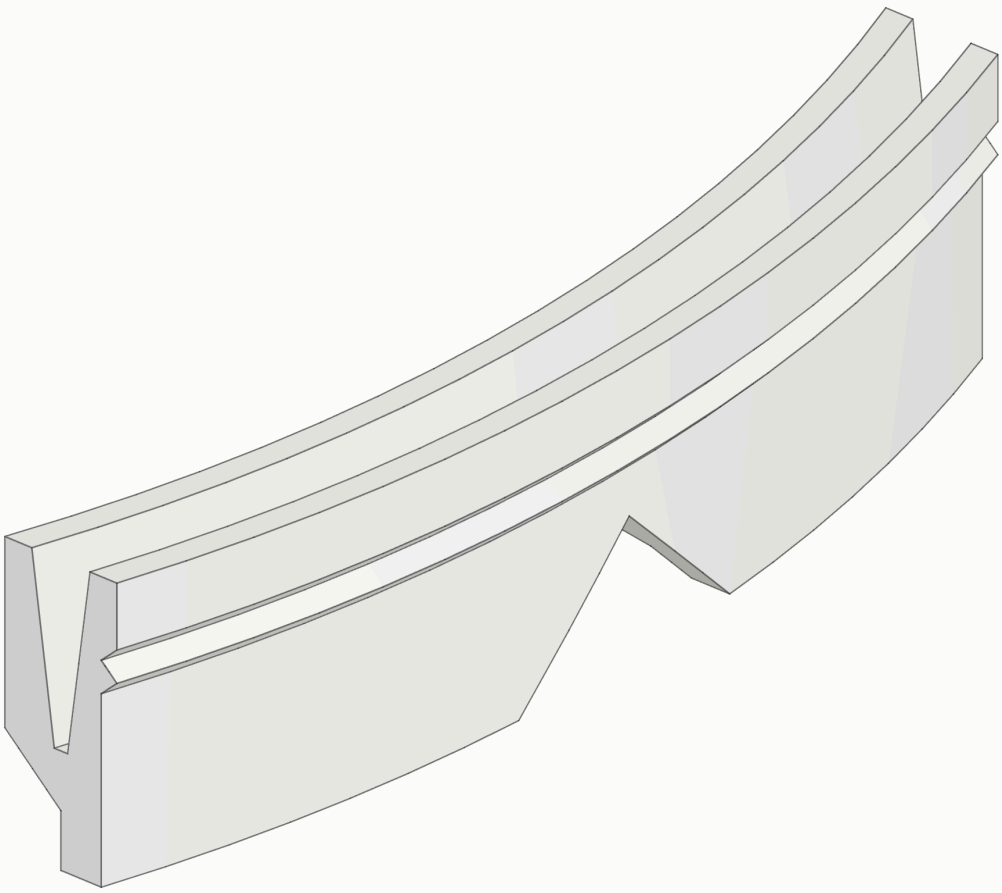
cutting ring | loaded cartridge | M2.5 x 12 screw | 2 mm hex key | jug | flat table



1. Empty the jug. Rinse it with water and let it dry.
2. One time only: put the insert on the hole in the shelf. Push it in with the soldering iron at 220 °C until it is flush with the shelf top. Let it cool.
3. Stand the cutting ring on a flat table. Put the jug down into the ring from above.
4. Put the loaded cartridge on the shelf, behind the ring, edge toward the jug. The side cheeks go down beside the shelf.
5. Push the back end of the cartridge with your thumb. The tip of the angled blade goes through the wall first.
6. Put the screw through the cartridge into the insert. Tighten it until the cartridge is tight on the shelf. The screw clamps the cartridge, and its shank stops the blade. You do not have to hold the cartridge.
7. Turn the jug by its handle CLOCKWISE (seen from above), one full turn. The arrow on the cartridge shows the way the near side of the jug moves. The blade cuts a level line.
8. Remove the screw. Pull the cartridge back. Lift off the top half, then lift the jug out.

3 Fit test

`fit test arc` `cut jug`

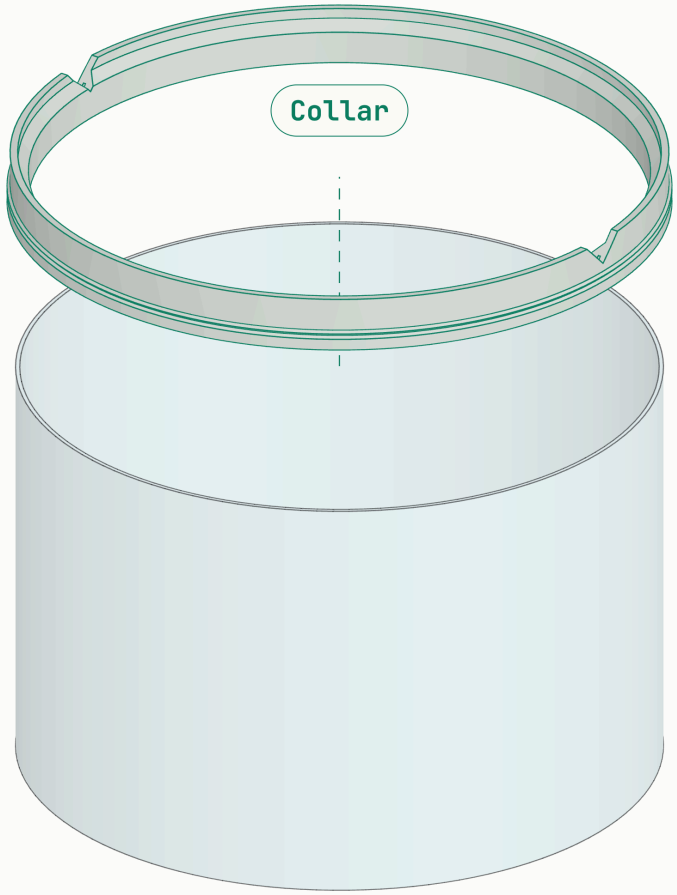


1. Push the groove of the fit test arc down over the cut edge.
2. The edge must go into the groove and stop with a firm push.
3. Look at the outer face. The skirt must follow the jug wall, with no gap at the ends of the arc.

Note. The jug diameter comes from one sample jug. Other jugs can differ. A gap at the ends: the diameter in the model is too large. A gap at the middle: it is too small. Measure the jug (`MEASURE.md`), change `JUG_DIA` in `shape.py`, run `make.sh`, and print the arc again.

4 Put on the collar

collar jug

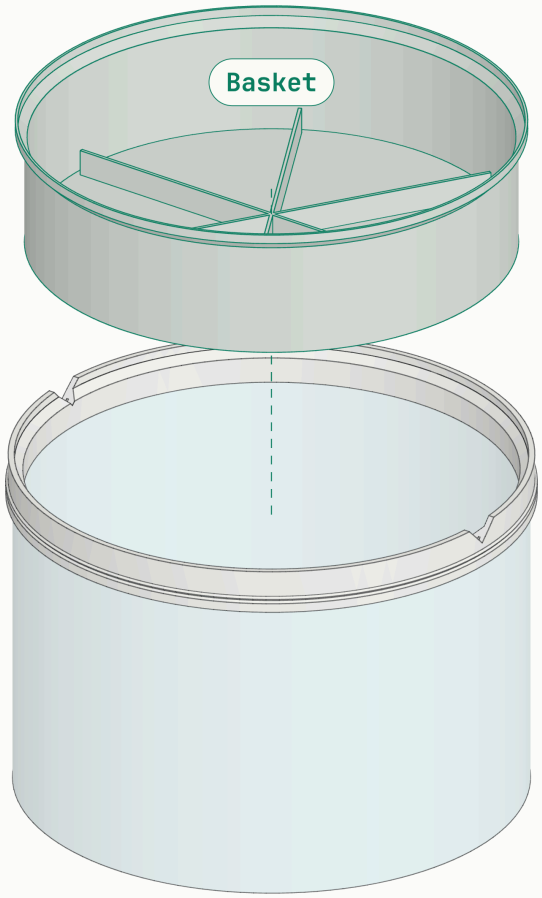


1. Turn the collar so that the groove opens down.
2. Put the groove over the cut edge of the jug. Start at one side and work around.
3. Push down hard all around. The edge wedges into the groove.
4. Pull up on the collar. It must stay on the jug.

Note. The groove tapers from 2.6 mm at its mouth to 0.6 mm at its roof. A wall from 0.6 to 2.4 mm wedges in. The lid snap also presses the collar skirt onto the jug.

5 Put in the basket

basket | desiccant pellets

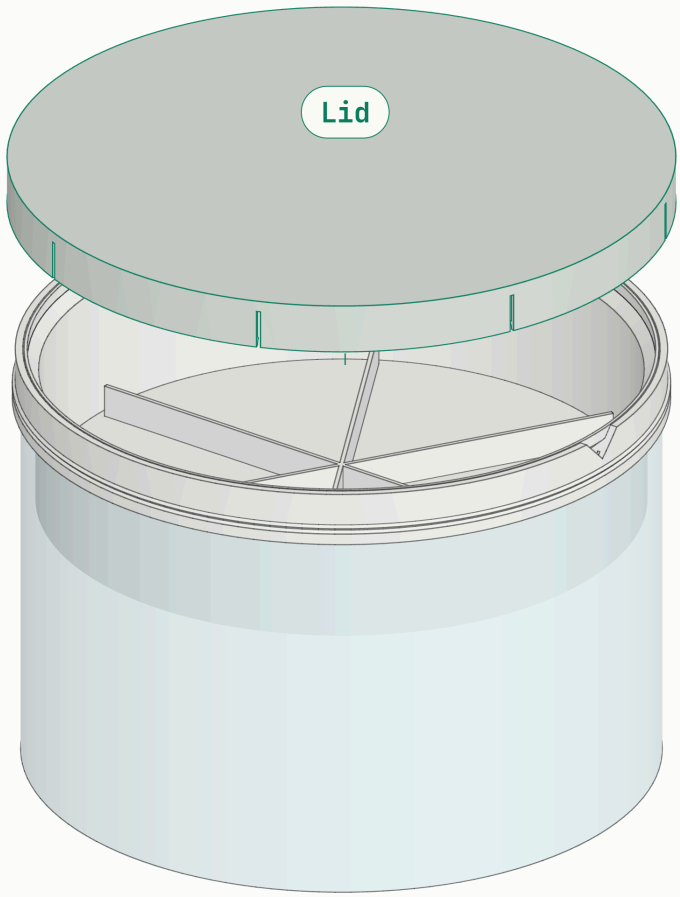


1. Lower the basket into the collar. Its flared rim sits on the 45° seat of the collar.
2. Fill the basket with 400 g (0.9 lb) of calcium chloride pellets: about 450 mL (1.9 US cups), 28 mm deep. This is 11 mm below the rim.
3. Do not fill it full. A full basket holds 0.63 L, about 560 g. At 70 % humidity, 560 g of pellets makes about 1.6 L of brine. The jug holds only 1.13 L of brine below the basket.
4. Use calcium chloride pellets, for example DampRid refills. The pellets must be larger than the 2 mm openings in the floor.
5. To lift the basket out, put your fingers into the two notches in the collar and pull up on the rim.

Note. The pellets take water from the air and turn to brine. The brine drips through the open grid of the floor into the jug.

6 Snap on the lid

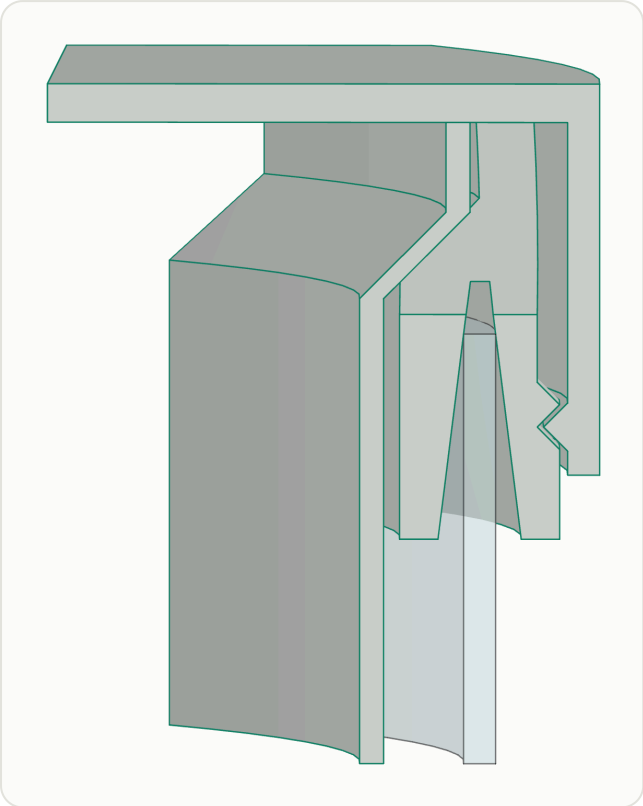
`lid`



1. Put the lid on the collar, with the pull tab at one side.
2. Push down on each side of the lid until it clicks. The bead of the skirt goes over the ridge of the collar.
3. To remove the lid, pull the tab out and up.
4. Empty the brine when it gets near the basket floor. Remove the collar with the basket and the lid. Pour the brine down a drain with water.

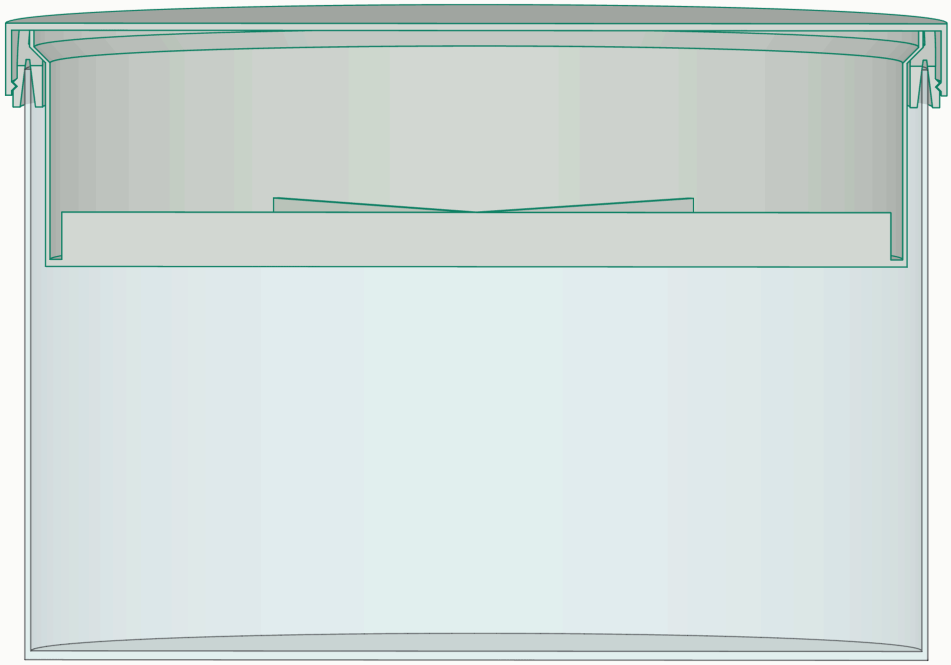
Note. The openings in the lid grid are 2.0 mm square. Air goes in, and the pellets stay in if the jug falls over. Brine can come out if the jug falls over: put the jug in a place where it can not fall.

How it locks



The section through the rim, at one side of the jug. From the left: the basket rim under the lid plate, the collar, the jug wall in the groove, and the lid skirt.

- **Groove.** The cut edge goes up into a tapered groove in the collar. A 1.0 mm wall stops 1.6 mm below the roof. A firm push wedges it in.
- **Seat.** The basket rim flares out at 45°. It sits on a 45° seat in the collar and centres itself.
- **Snap.** A bead inside the lid skirt goes over a ridge on the collar skirt, and clicks into a groove. The snap is 0.5 mm. 8 slits cut the skirt into fingers. Each finger bends with 1.17 % strain (PLA takes about 1.5 %).
- **Clamp.** The lid skirt holds the collar skirt against the jug wall.
- **Rim.** The lid plate sits on the basket rim. The pellets stay in.



The section through the middle. The basket holds 0.6 L of pellets. The space below the basket holds 1.1 L of brine (cut at 100 mm). A star of ribs (6 spokes) stiffens the floor. The floor prints as an open grid of connected lines, 2.5 mm apart: one path for each layer, with no small holes.

Adjust the fit

The jug wall is an estimate. Measure the jug (`MEASURE.md`), change the values in `shape.py`, run `make.sh`, and print the fit test arc again.

Problem	Value	Now (mm)	Change
The collar is loose on the jug, or it does not go over the edge	`JUG_DIA`	153	Measure the circumference of the jug at the cut, and divide by 3.1416 (MEASURE.md, J1).
The blade does not cut through the wall, or it tears it	`YAW`	30	Turn the jug clockwise. Change YAW by 5: a larger angle goes deeper, a smaller angle slices more.
The jug does not go into the cutting ring, or it is loose in it	`RING_CLR`, `JUG_DIA`	0.6, 152.8	Check JUG_DIA (MEASURE.md, J1). Change RING_CLR by 0.2.
The edge does not come out of the front, and the screw does not go in	`SLOT_CLR`, `POCKET_CLR`	0.75, 0.50	The slot printed closed. Do not force the screw. Add 0.2 to SLOT_CLR, or 0.2 to POCKET_CLR if the spine sticks.
The insert is loose in the shelf, or it does not go in	`INSERT_D`	3.6	Set INSERT_D to the hole size on the data sheet of your insert. Reprint the ring.
The blade is loose in the slot	`SLOT_CLR`	0.75	Subtract 0.1 from SLOT_CLR.
The edge stops at the mouth of the groove	`JUG_T`, `G_HALF_MOUTH`	1.0, 1.3	Measure the wall (J2). Set JUG_T to the value. For a wall over 2.2 mm, add 0.3 to G_HALF_MOUTH.
The collar is loose on a correct jug size	`G_HALF_ROOF`	0.3	Subtract 0.1 from G_HALF_ROOF. The wall then wedges higher in the groove.
The lid is too hard to put on, or it is loose	`SNAP`	0.50	Subtract or add 0.1. The skirt strain must stay below 1.5 % (model.py prints it).
The basket sits too high or too low	`DEPTH, CUT_H`	35, 100	Change DEPTH for a deeper or shallower basket. CUT_H is only for the brine volume and the images.

The fit check. `model.py` checks the collar on the jug, the basket in the collar, and the lid on the collar, also on the path in. It stops if a part touches where it must not, if the collar has no grip, or if the lid has no snap.